

Mathematical Statistics I
HOMEWORK 7

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1.

$$\mu = 400$$

$$\sigma = 20$$

a.

$$P(X > 450) = P\left(Z > \frac{450 - 400}{20}\right) = P(Z > 2.5) = 0.0062$$

$$P(X > 450) = 0.0062$$

There is a 0.62% probability the actual cost will exceed the \$450 budget.

b.

$$P(X > x) = 0.10$$

** value closest to 0.10 is 0.1003**

$$P(Z > 1.28) = 0.10$$

$$Z = \frac{x - \mu}{\sigma}$$

$$1.28 = \frac{x - 400}{20}$$

$$1.28 \cdot 20 = x - 400$$

$$25.6 = x - 400$$

$$x = 425.6$$

The company should budget \$425.60.

2.

$$\mu = 0.13$$

$$\sigma = 0.005$$

a.

$$P(0.12 \leq X \leq 0.14)$$

$$P\left(\frac{0.12 - 0.13}{0.005} \leq Z \leq \frac{0.14 - 0.13}{0.005}\right) = P\left(\frac{-0.01}{0.005} \leq Z \leq \frac{0.01}{0.005}\right) = P(-2 \leq Z \leq 2)$$

$$P(Z > 2) = 0.0228$$

$$P(Z > -2) = 1 - 0.0228 = 0.9772$$

$$P(Z < 2) - P(Z < -2) = 0.9772 - 0.0228 = 0.9544$$

$$P(\text{a random wire meets specification}) = 0.9544$$

The probability that a randomly selected wire from company A's production will meet the specifications is 95.44%.

b.

$$P(\text{a random wire meets specification}) = 0.9544$$

$$P(\text{all 4 meet specification}) = (P(\text{one meets specification}))^4$$

$$P(\text{all 4 meet specification}) = (0.9544)^4 \sim 0.8297$$

$$P(\text{all 4 meet specification}) \sim 0.8297$$

The probability that all four in a randomly selected system will meet the specifications is approximately 82.97%.

3.

a.

$$\mu = ?$$

$$\sigma = 0.3$$

$$P(X > 8) = 0.01$$

$$P\left(Z > \frac{8 - \mu}{0.3}\right) = 0.01$$

** value closest to 0.01 is 0.0099**

$$P(Z > 2.33) = 0.01$$

$$\frac{8 - \mu}{0.3} = 2.33$$

$$8 - \mu = 2.33 \cdot 0.3$$

$$8 - \mu = 0.699$$

$$\mu = 8 - 0.699 = 7.301$$

$$\mu \sim 7.30 \text{ ounces}$$

b.

$$P(\mu - 1 \leq X \leq \mu + 1) = 0.95$$

$$P(-1 \leq X - \mu \leq 1) = P\left(-\frac{1}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{1}{\sigma}\right)$$

$$P\left(-\frac{1}{\sigma} \leq Z \leq \frac{1}{\sigma}\right) = 0.95$$

95% is in the middle, 5% total in the tails, therefore each tail has 0.025.

$$P(Z > 1.96) = 0.0250$$

$$\frac{1}{\sigma} = 1.96$$

$$\sigma = \frac{1}{1.96} = 0.5102$$

$$\sigma = 0.51$$

4.

$$X \sim \text{Exponential}(\mu = 2.4)$$

a.

$$P(X > 3.0) = 1 - F(3.0) = e^{\frac{-3.0}{2.4}} = e^{-1.25} \sim 0.2865$$

$$P(X > 3.0) \sim 0.2865$$

The probability the earthquake striking this region will exceed 3.0 is 28.65%

b.

$$P(2 \leq X \leq 3) = P(X > 2) - P(X > 3) = e^{\frac{-2}{\mu}} - e^{\frac{-3}{\mu}} = e^{\frac{-2}{2.4}} - e^{\frac{-3}{2.4}}$$

$$P(2 \leq X \leq 3) = e^{-0.8333} - e^{-1.25} = 0.4346 - 0.2865 = 0.1481$$

$$P(2 \leq X \leq 3) = 0.1481$$

The probability the earthquake falls between 2.0 and 3.0 is 14.81%.

c.

$$P(X > 5) = 1 - F(5) = e^{\frac{-5}{2.4}} = e^{-2.0833} \sim 0.1242$$

$$P(\text{earthquake does not exceed } 5.0) = 1 - p = 1 - 0.1242 = 0.8758$$

$$P(\text{next 10 earthquakes exceed } 5.0) = (1 - p)^{10} = 0.8758^{10} \sim 0.2634$$

$$P(\text{at least one will exceed } 5.0) = 1 - 0.2634 = 0.7366$$

$$P(\text{at least one will exceed } 5.0) = 0.7366$$

The probability that at least one will exceed 5.0 is 73.66%.

5.

$$\begin{aligned}\mu_R &= 10 \\ \lambda &= \frac{1}{\mu_R} = \frac{1}{10} = 0.1 \\ A &= \pi R^2\end{aligned}$$

$$\begin{aligned}E[R] &= \mu_R = \frac{1}{\lambda} \\ \text{Var}(R) &= \frac{1}{\lambda^2}\end{aligned}$$

$$E[R^2] = \text{Var}(R) + [E(R)]^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = \frac{2}{\lambda^2}$$

$$E[A] = E[\pi R^2] = \pi E[R^2] = \pi \left(\frac{2}{\lambda^2} \right) = \frac{2\pi}{\lambda^2} = \frac{2\pi}{(0.1)^2} = 2\pi(100) = 200\pi$$

$$E[A] = 200\pi \sim 628.32$$

$$E[A] \sim 628.32$$

$$\text{Var}(A) = \pi^2 \text{Var}(R^2)$$

$$\text{Var}(R^2) = E[R^4] - [E(R^2)]^2$$

$$\text{Var}(R^2) = \frac{24}{\lambda^4} - \left(\frac{2}{\lambda^2} \right)^2 = \frac{20}{\lambda^4}$$

$$\text{Var}(A) = \pi^2 \cdot \frac{20}{\lambda^4} = \frac{20\pi^2}{\lambda^4}$$

$$\text{Var}(A) = \frac{20\pi^2}{(0.1)^4} = 200,000\pi^2$$

$$\text{Var}(A) \sim 1.97 \times 10^6$$